

Certified Penetration Testing Professional

Methodology: Network Penetration Testing - Perimeter Devices

Penetration Tester:			
Organization:			
Date:		**Location:**	

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Test 1: Assessing Firewall Security Implementation

Test 1.1: Test the Firewall from Both Sides

Target Organization		
URL		
Firewall Tested Successfully?	☐ Yes	☐ No
Activities Performed for Testing the Firewall from Inside	1. 2. 3. 4. 5.	
Activities Performed for Testing the Firewall from Outside	1. 2. 3. 4. 5.	
Identified Firewall Rules	1. 2. 3. 4. 5.	
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.2: Find Information about the Firewall

Target Organization	
URL	
Information Available	
Company's Name Server	
Topographic Information	
Target IP Address	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 1.3: Locate the Firewall by Conducting Traceroute

Target Organization	
URL	
Firewall Location	
Firewall IP Address	
Network Topology	
Routers	
Filtering Devices	
Protocols Allowed	
Protocols Denied	

IP Addresses Hoped
1.
2.
3.
4.
5.
6.
7.
8.
9.
10.
11.
12.
13.
14.
15.
16.
17.

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18.	
19.	
20.	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 1.4: Try to Pass through the Firewall using Hping

Target Organization		
URL		
Command used		
IP of Tested Firewall		
Custom Packets Created?	☐ Yes	☐ No
Response Received		
Packet Crafter used		
Successfully Pass-Through Firewall using Hping?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.5: Enumerate Firewall Access Control List using Nmap

Target Organization				
URL				
Command used				
Scanning methods	State of ports			
SYN stealth scan	☐ Open	☐ Filtered	☐ Unfiltered	☐ Closed
SYN stealth scan through Firewall	☐ Open	☐ Filtered	☐ Unfiltered	☐ Closed
Source ports scan	☐ Open	☐ Filtered	☐ Unfiltered	☐ Closed
ACK scan	☐ Open	☐ Filtered	☐ Unfiltered	☐ Closed
FIN scan	☐ Open	☐ Filtered	☐ Unfiltered	☐ Closed
Null Scan	☐ Open	☐ Filtered	☐ Unfiltered	☐ Closed
Win packet Scan	☐ Open	☐ Filtered	☐ Unfiltered	☐ Closed
XMAS scan	☐ Open	☐ Filtered	☐ Unfiltered	☐ Closed
UDP scan	☐ Open	☐ Filtered	☐ Unfiltered	☐ Closed
Response received				
Tools/Services Used	1. _____ 2. _____ 3. _____ 4. _____ 5. _____			

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Test 1.6: Scan the Firewall for Vulnerabilities

Target Organization			
URL			
Firewall Scanning Successful?		☐ Yes	☐ No
Vulnerabilities Identified			
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.7: Map Firewall Make and Version with Associated Vulnerabilities

Target Organization		
URL		
Firewall is Patched Up?	☐ Yes	☐ No
Exploit Obtained in Google Hacking Database		
Response Received		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.8: Try to Bypass the Firewall using Fragmented Packets

Target Organization			
URL			
IP of Tested Firewall			
Command used			
MTU Number Specified			
Successfully bypassed the firewall using fragmented packets?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.9: Try to Bypass the Firewall by Spoofing Packets

Target Organization		
URL		
IP of Tested Firewall		
Command used		
Successfully bypassed firewall by spoofing packets?	☐ Yes	☐ No
IP Addresses of Decoys		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.10: Try to Bypass Firewall by Spoofed Source Port

Target Organization			
URL			
IP of Tested Firewall			
Command used			
Source port number used			
Successfully bypassed the firewall by the spoofed source port?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.11: Try to Bypass Firewall by MAC Address Spoofing

Target Organization			
URL			
IP of Tested Firewall			
Command used			
Successfully bypassed the firewall by MAC address spoofing?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.12: Try to Bypass the Firewall by IP Address Spoofing

Target Organization			
URL			
Command used			
IP of Tested Firewall			
Modified Addressing Information			
Successfully bypassed the firewall by IP address spoofing?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.13: Try to Bypass the Firewall by Varying Packet Size

Target Organization			
URL			
IP of Tested Firewall			
Command used			
Packet size used			
Successfully bypassed the firewall by varying packet size?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.14: Try to Bypass the Firewall by Sending Bad Checksums

Target Organization		
URL		
IP of Tested Firewall		
Command used		
Successfully bypassed the firewall by sending bad checksums?	☐ Yes	☐ No
Is System well configured?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.15: Try to Bypass the Firewall using Port Redirection

Target Organization		
URL		
IP of Tested Firewall		
Command used		
IP used		
Port redirector installed successfully?	☐ Yes	☐ No
Successfully bypassed the firewall using port redirection?	☐ Yes	☐ No
Selected port number		
Response received		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.16: Try to Bypass the Firewall Using IP Address in Place of URL

Target Organization			
URL			
IP of Tested Firewall			
Command used			
Online services used			
IP address used			
Successfully bypassed the firewall using the IP address in place of URL?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.17: Try to Bypass the Firewall Using Anonymous Website Surfing Sites

Target Organization			
URL			
IP of Tested Firewall			
IP address			
Successfully bypassed the firewall using anonymous website surfing sites?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.18: Try to Bypass the Firewall using a Proxy Server

Target Organization		
URL		
IP of Tested Firewall		
Proxy server used		
IP address used		
Port number used		
Successfully bypassed the firewall using a proxy server?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.19: Try to Bypass the Firewall using Source Routing

Target Organization		
URL		
IP of Tested Firewall		
Successfully bypassed the firewall using source routing?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.20: Try to Bypass the Firewall using HTTP Tunneling Method

Target Organization			
URL			
IP of Tested Firewall			
Target company			
Information required			
Successfully bypassed the firewall using HTTP tunneling method?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.21: Try to Bypass the Firewall using ICMP Tunneling Method

Target Organization			
URL			
IP of Tested Firewall			
ICMP Tunneling Technique Results			
Successfully bypassed the firewall using the ICMP tunneling method?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.22: Try to Bypass the Firewall using ACK Tunneling Method

Target Organization			
URL			
IP of Tested Firewall			
ACK Tunneling Technique Results			
Successfully bypassed the firewall using the ACK tunneling method?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 1.23: Try to Bypass the Firewall using SSH Tunneling Method

Target Organization		
URL		
IP of Tested Firewall		
SSH Tunneling Technique Results		
Successfully bypassed the firewall using SSH tunneling method?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.24: Try to Bypass the Firewall through MITM Attack

Target Organization			
URL			
IP of Tested Firewall			
DNS server			
Successfully bypassed the firewall through the MITM attack?		☐ Yes	☐ No
Tools/Services Used	1. _____ 2. _____ 3. _____ 4. _____ 5. _____		

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Test 1.25: Try to Bypass the Firewall using Malicious Contents

Target Organization			
URL			
IP of Tested Firewall			
Malicious Content Used against Firewall			
Successfully bypassed the firewall Using malicious contents?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2: Assessing IDS Security Implementation

Test 2.1: Test for Resource Exhaustion

Target Organization	
URL	
IDS Network Tested	
IDS Limitations	
IDS Performance	
Point of Resource Exhaustion	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.2: Test the IDS by Sending an ARP Flood

Target Organization			
URL			
IDS Network Tested			
Performed Network Flooding by sending ARP Packets Successfully?		☐ Yes	☐ No
IDS Response			
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.3: Test the IDS by MAC Spoofing

Target Organization			
URL			
IDS Network Tested			
Spoofed MAC Address			
Sent spoofed MAC address to the IDS Successfully?		☐ Yes	☐ No
IDS Response			
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.4: Test the IDS by IP Spoofing

Target Organization		
URL		
IDS Network Tested		
Spoofed IP Address		
Sent Spoofed IP Address Successfully?	☐ Yes	☐ No
Response Received		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2.5: Test the IDS by Sending SYN Floods

Target Organization			
URL			
Tested IDS by Sending SYN Floods?		☐ Yes	☐ No
Results from the IDS Test Performed	1. 2. 3. 4. 5.		
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.6: Test the IDS by Editing and Replaying Captured Network Traffic

Target Organization	
URL	
IDS Network Tested	
Attempted Packets	
Successful Packets	
Truncated Packets	
Retried Packets	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.7: Test the IDS for a Denial-of-Service (DoS) Attack

Target Organization		
URL		
IDS Network Tested		
Does IDS have central logging servers?	☐ Yes	☐ No
Response of the DoS attack on IDS		
Central log server's IP address		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2.8: Try to Bypass IDS using Anonymous Website Surfing Sites and a Proxy Server

Target Organization	
URL	
IDS Network Tested	
Anonymous Web-Surfing Sites	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.9: Try to Bypass IDS using Botnet

Target Organization		
URL		
Botnet Detection Successful?	☐ Yes	☐ No
Approach used For Botnet Detection		
IDS Network Tested		
Response of the DoS Attack on IDS		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2.10: Test the IDS by Sending Inconsistent Packets

Target Organization		
URL		
IDS Network Tested		
Inconsistent Packets		
Sent Inconsistent TCP/IP or UDP/IP Packets with different TCP/UDP and IP Header Sizes Successfully?	☐ Yes	☐ No
IDS Response		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2.11: Test the IDS for IP Packet Fragmentation

Target Organization		
URL		
IDS Network Tested		
Packet Size Sent		
Sent IP Packet Fragments Successfully?	☐ Yes	☐ No
IDS Response		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2.12: Test the IDS for Polymorphic Shellcode

Target Organization	
URL	
IDS Network Tested	
Response of the IDS against the Polymorphic Shellcode	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.13: Try to Evade the IDS by Obfuscating or Encoding the Attack Payload

Target Organization	
URL	
IDS Network Tested	
Obfuscated code	
Response of IDS	
Response of Target System	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.14: Check the IDS for False-Positive Generation

Target Organization			
URL			
IDS Network Tested			
Does IDS activate a large number of false reports?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.15: Test the IDS for TTL Evasion

Target Organization	
URL	
IDS Network Tested	
How was TTL evasion attempted?	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.16: Test the IDS by Sending a Packet to Port 0

Target Organization	
URL	
IDS Network Tested	
Packet sent to Port 0	
Response Received	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.17: Test the IDS for UDP Checksum

Target Organization	
URL	
IDS Network Tested	
UDP Checksum Information	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.18: Test the IDS for TCP Retransmissions

Target Organization	
URL	
IDS Network Tested	
Retransmitted TCP Packets	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.19: Test the IDS using Covert Channels

Target Organization		
URL		
Tested IDS Using Covert Channels?	☐ Yes	☐ No
Information Gathered Using Covert Channels	1. 2. 3. 4. 5.	
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2.20: Test the IDS for Reverse Traversal

Target Organization			
URL			
Tested the IDS for Reverse Traversal?		☐ Yes	☐ No
Findings from the Test	1. 2. 3. 4. 5.		
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.21: Test for Self-Referencing Directories

Target Organization		
URL		
Tested the IDS for Self-Referencing Directories?	☐ Yes	☐ No
Findings from the Test	1. 2. 3. 4. 5.	
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2.22: Test for Premature Request Ending

Target Organization			
URL			
Tested the IDS for Premature Request Ending?		☐ Yes	☐ No
Findings from the Test	1. 2. 3. 4. 5.		
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.23: Test for the IDS Parameter Hiding

Target Organization			
URL			
Tested for IDS Parameter Hiding?		☐ Yes	☐ No
Findings from the Test	1. 2. 3. 4. 5.		
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.24: Test the IDS for HTTP Misformatting

Target Organization		
URL		
Tested IDS for HTTP Misformatting?	☐ Yes	☐ No
Findings from the Test	1. 2. 3. 4. 5.	
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2.25: Test the IDS for Long URLs

Target Organization			
URL			
Tested IDS for Long URLs?		☐ Yes	☐ No
Findings from the Test	1. 2. 3. 4. 5.		
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.26: Test for Null Method Processing

Target Organization			
URL			
Tested IDS for Null Method Processing?		☐ Yes	☐ No
Findings from the Test	1. 2. 3. 4. 5.		
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.27: Try to Bypass the IDS using Compressed Media Files

Target Organization		
URL		
Can IDS identify the attack code within the compressed data?	☐ Yes	☐ No
Findings from the Test	1. 2. 3. 4. 5.	
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2.28: Test Session Splicing

Target Organization			
URL			
Tested IDS for Session Splicing?		☐ Yes	☐ No
Findings from the Test	1. 2. 3. 4. 5.		
Sessions Susceptible to Malicious Data	1. 2. 3. 4. 5.		
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 2.29: Try to Bypass Invalid RST Packets through the IDS

Target Organization	
URL	
Expected Checksum	
Received Checksum	
Examined Two-way Communication TCP Protocols Using RST Packets	
Results from Verified RST Packets and Checksum	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 3: Assessing the Security of Routers

Test 3.1: Identify the Router Hostname

Target Organization	
URL	
IP address of the router	
Hostname of the router	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 3.2: Port Scan the Router

Target Organization	
URL	

Open Ports	
☐ **7** Echo	☐ **113** IDENT
☐ **13** DayTime	☐ **115** Simple File Transfer Protocol (SFTP)
☐ **17** Quote of the Day (QOTD)	☐ **137** NetBIOS
☐ **20** File Transfer Protocol (FTP)	☐ **138** NetBIOS
☐ **21** File Transfer Protocol (FTP)	☐ **139** NetBIOS
☐ **22** Secure Socket Shell (SSH)	☐ **143** Internet Message Access Protocol (IMAP)
☐ **23** Telnet	☐ **161** Simple Network Management Protocol
☐ **25** SMTP	☐ **162** Simple Network Management Protocol
☐ **53** Domain Name System (DNS)	☐ **194** Internet Relay Chat (IRC)
☐ **63** Whois	☐ **443** HTTPS
☐ **66** SQL*net (Oracle)	Other Ports:
☐ **70** Gopher	
☐ **79** Finger	
☐ **80** HTTP	
☐ **88** Kerberos	
☐ **101** Host Name Server	
☐ **109** Post Office Protocol 2 (POP2)	
☐ **110** Post Office Protocol 3 (POP3)	

Tools/Services Used	1. 2. 3. 4. 5.

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Test 3.3: Identify the Router Operating System and its Version

Target Organization	
URL	
IP address of the router tested	
Operating System and its version	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 3.4: Identify Protocols Running

Target Organization		
URL		
IP address of the router tested		
Protocols running		
☐ RIP	☐ OSPF	
☐ RIPv2	☐ BGP	
☐ IGRP	☐ Others	
☐ EIGRP		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.5: Test for TFTP Connections

Target Organization		
URL		
IP address of the router tested		
TFTP Allowed?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.6: Try to Retrieve the Router Configuration File

Target Organization			
URL			
Command used			
IP used			
Sniffed TFTP traffic from the wire successfully to retrieve the router configuration file?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.7: Test for Router Misconfigurations

Target Organization		
URL		
IP address of the router tested		
Is the router misconfigured?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.8: Try to Recover Router Passwords from Config File

Target Organization			
URL			
Input information to crack Router Password			
Router IP Address			
Router Username			
Router Password			
Successfully recovered router passwords from config file?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.9: Test for VTY/TTY Connections

Target Organization			
URL			
IP address of the router tested			
Is console access possible?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.10: Try to Gain Access to the Router

Target Organization			
URL			
Command used			
IP address used			
Standard port used			
Is the modem connected to the device?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.11: Test for Router Running Modes

Target Organization		
URL		
IP address of the router tested		
Modes	☐ USER MODE	☐ PRIVILEGE MODE
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.12: Privileged Mode Attacks

Target Organization			
URL			
Command used			
Response received			
Performed brute-force password attacks to crack the password?		☐ Yes	☐ No
Is the password configured?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.13: Test for SNMP Capabilities

Target Organization	
URL	
IP address of the router tested	
SNMP Strings used	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 3.14: Perform SNMP Bruteforcing

Target Organization		
URL		
Successfully performed SNMP brute-forcing?	☐ Yes	☐ No
SNMP community strings		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.15: Try to Log in using the default SNMP Community String

Target Organization			
URL			
Is default SNMP community string "public" changed at?		☐ Yes	☐ No
Successfully logged in using default SNMP community string?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.16: Test if Finger is Running on the Router

Target Organization			
URL			
IP address of the router tested			
Finger Service running?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.17: Test for CDP Protocol Running on the Router

Target Organization			
URL			
IP address of the router tested			
CDP Protocol running?		☐ Yes	☐ No
CDP Messages			
☐ Device ID (hostname)		☐ IOS software version being used	
☐ Port ID (port information about the sender)		☐ Capabilities of the router	
☐ Operating system platform		☐ Network IP address	
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.18: Test for NTP Protocol

Target Organization			
URL			
IP address of the router tested			
NTP Protocol running?		☐ Yes	☐ No
Router Synchronized?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.19: Test for Loose and Strict Source Routing

Target Organization		
URL		
IP address of the router tested		
Routing	☐ Loose Source Routing	☐ Strict Source Routing
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.20: Test for IP Spoofing

Target Organization		
URL		
IP address of the router tested		
IP Spoofing possible?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.21: Test for IP Handling Bugs

Target Organization			
URL			
IP address used			
Is ICMP redirects manipulated the host?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.22: Test ARP Attacks

Target Organization		
URL		
IP address of the router tested		
ARP spoofing is possible against the router?	☐ Yes	☐ No
Victim IP address		
Victim MAC address		
Poisoned IP address		
Poisoned MAC address		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.23: Test BGP protocol

Target Organization			
URL			
IP address of the router tested			
BGP Protocol present?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.24: Test for EIGRP protocol

Target Organization			
URL			
IP address of the router tested			
EIGRP Protocol present?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.25: Test Router Denial-of-Service Attacks

Target Organization			
URL			
IP address of the router tested			
Malformed Packet Attack?		☐ Yes	☐ No
Packet Flood Attacks?		☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 3.26: Test Router's HTTP Capabilities

Target Organization	
URL	
IP address of the router tested	
Port Used to Connect	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 3.27: Test for HTTP Configuration Vulnerabilities in Cisco Routers

Target Organization		
URL		
Gained full remote administrative access on Cisco devices?	☐ Yes	☐ No
Response received		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.28: Test through HSRP Attack

Target Organization		
URL		
IP address of the router tested		
HSRP group forwarded to IP address?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 3.29: Router Penetration Testing using Secure Cisco Auditor (SCA)

Target Organization			
URL			
IP address of the router tested			
Routing Penetration Testing Successful?		☐ Yes	☐ No
ID	**Risk Level**	**Category**	**Name**
1.			
2.			
3.			
4.			
5.			
Tools/Services Used	1. 2. 3. 4. 5.		

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Test 4: Assessing Security of Switches

Test 4.1: Look for Security Misconfigurations in Cisco Switch Configuration

Target Organization					
URL					
The common switch security misconfiguration checks for CISCO and other Manufacturers	**Default vulnerable configurations**	**Unused ports**	**DHCP snooping**	**Port security**	**Correct timestamp**
Tools/Services Used	1.				
	2.				
	3.				
	4.				
	5.				

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Test 4.2: Test for Address of Cache Size

Target Organization	
URL	
Frame size relayed	
Address Cache Size	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.3: Test for Data Integrity and Error Checking

Target Organization	
URL	
Frame Size	
Traffic Rate	
Data Pattern	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.4: Test for Back-to-Back Frame Capacity

Target Organization	
URL	
Number of frames sent at once	
Inter-frame gaps	
Number of frames forwarded by the switch	
Number of test rerun	
Capacity detected	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.5: Test for Frame Loss

Target Organization	
URL	
Count the frames that are transmitted	
Frame loss equation	
Measurement	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.6: Test for Latency

Target Organization	
URL	
Method used	
Latency detected	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.7: Test for Throughput

Target Organization	
URL	
Count the frames	
The rate of the offered stream	
Throughput	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.8: Test for Frame Error Filtering

Target Organization	
URL	
Frame Size	
Illegal frame types	
Traffic Rate	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.9: Test For Fully Meshed Condition

Target Organization	
URL	
Frame Size	
Traffic Rate	
Traffic Data Type	
DUT setup	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.10: Functional Test for Stateless QoS

Target Organization	
URL	
Frame size	
Duration	
Traffic Rate	
DUT-QoS	
DUT-Line speed	
DUT-QoS type	
DUT-QoS Policies	
DUT-Queue type	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.11: Performance Test for Spanning Tree Network Convergence

Target Organization	
URL	
Test ports	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.12: Test for OSPF Performance

Target Organization	
URL	
Frame Size	
Traffic Rate	
OSPF Parameters	
DUT setup	
DUT OSPF Area	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.13: Test for VLAN Hopping

Target Organization	
URL	
Dynamic Trunking Protocol	
DTP States	
DTP Negotiation	
VLAN Hopping	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.14: Test for MAC Table Flooding

Target Organization	
URL	
Content Addressable Memory	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.15: Testing for ARP Attack

Target Organization	
URL	
MAC address	
IP address	
Tools/Services Used	1. _____ 2. _____ 3. _____ 4. _____ 5. _____

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Test 4.16: Check for VTP Attack

Target Organization	
URL	
Command Output	
Cat2950#show vtp status	
VTP Version	
Configuration Revision	
Maximum VLANs supported locally	
Number of existing VLANs	
VTP Operating Mode	
VTP Domain Name	
VTP Pruning Mode	
VTP V2 Mode	
VTP Traps Generation	
MD5 digest	
Configuration last modified by	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 4.17: Router and Switch Security Auditing using Traffic IQ Professional

Target Organization		
URL		
Security Auditing Successful?	☐ Yes	☐ No
Number of Network Switches Identified		
Number of Routers Identified		
Systems Assessed		
Tools/Services Used	1. 2. 3. 4. 5.	

Results Analysis:

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